

White paper

5th Gen Intel® Xeon® Scalable Processors
Intel® Memory Resilience Technology (Intel® MRT)

The Intel logo is displayed in white text on a blue square background.

5th Gen Intel® Xeon® Scalable Processor Empowering Alibaba Cloud to Maximize DDR5 Memory Reliability through AI-assisted Failure Analysis in BMC

A vertical illustration on the left side of the page shows a blue-tinted image of a server rack with glowing lights, overlaid with a large, stylized 'AI' logo and circuit-like patterns.

Background

Memory failure is among the leading causes of server failure in Alibaba Cloud datacenters. In large-scale datacenter, memory failure not only reduces server reliability and potentially disrupts datacenter continuity, but also impacts server performance. Memory reliability is a key element of server reliability, availability, and services (RAS) in the datacenters. Double Data Rate 5 (DDR5) is the new generation of memory with higher bandwidth, lower power, and higher density. However, it also introduces new challenges for memory reliability, including new memory architecture and signal design, denser capacity with higher risk of failure, and vaguer host error observation due to In-DRAM error correction code (ECC). Increasing reliability requirement from cloud service providers (CSP) further compound these challenges.

To address these issues, Alibaba Cloud and Intel have jointly improved DDR5 memory reliability by implementing unified out-of-band (OOB) memory error data collection and build-in artificial intelligence-assisted (AI-assisted) failure analysis in baseboard management controller (BMC). Intel® Memory Resilience Technology (Intel® MRT) is deployed in Alibaba Cloud fleets to provide early detection and prevention of potential memory failures. Memory health assessment results and predictive alerts are integrated with Alibaba Cloud Cruiser System (Alibaba Cloud's Server Monitoring System), which provides fast and comprehensive hardware monitoring services to ensure business stability in Alibaba Cloud datacenters.

The challenges of memory reliability

Memory fault can lead to a variety of memory errors, e.g., single bit errors (SBE), row type errors, column type errors, multi-array errors, bank errors, and DIMM errors. Each type of memory error has own frequency and victim pattern. Some error types are sporadic or intermittent, which makes them difficult to trace, while others can be repeatable. Some error types have higher risk of becoming uncorrectable errors (UE), which requires immediate RAS action, while others may have a lower risk of triggering UE but can lead to thousands of correctable errors (CE) in a short period of time, generating a CE storm that impacts system performance. There is no one-size-fits-all solution to address all memory errors.

Traditional solution is to replace the failed DIMM after the observation of UE or predict memory failure based on a counting-based CE rating policy. However, replacing DIMM after UE cannot avoid the cost of system crash, and CE-rating based UE prediction policy is far less effective in predicting complex memory failures. The occurrence of both CEs and UEs depends not only on the memory fault status of the hardware, but also on the implicit runtime context, ECC coverage, and memory-specific fault patterns. As a result, memory errors are highly uncertain, and predicting UEs becomes extremely difficult.

Maximizing DDR5 memory reliability through AI-assisted failure analysis in BMC

Alibaba Cloud and Intel have jointly researched and developed a memory failure prediction and prevention solution with unified out-of-band memory error data collection and built-in artificial intelligence-assisted failure analysis in the baseboard management controller (BMC). The solution is integrated with Alibaba Cloud Cruiser System, which provides fast and comprehensive hardware monitoring services to ensure business stability in Alibaba Cloud datacenters.

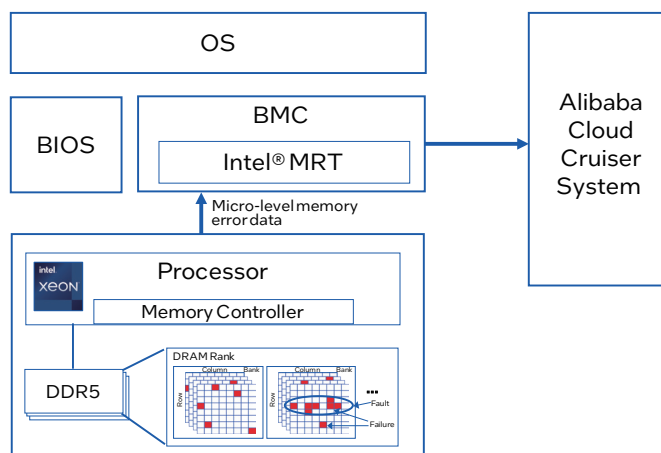


Figure 1. Solution Architecture

▪ Fine-grained memory error collection through BMC

The solution uses BMC to collect fine-grained correctable error (CE) and uncorrectable error (UE) information with detailed bit-level error information, instead of collecting memory error data using in-band such as error detection and correction (EDAC) driver or based upon BIOS triggered SMI. In-band EDAC driver cannot guarantee UE data collection and has the limitation of being unified deployed across diverse operational environments. CE threshold-based SMI is less useful in predicting memory failures. Out-of-band memory collection using BMC provides a reliable and unified memory error collection with fine-grained data granularity and rich error information.

▪ Micro-level memory fault aware failure detection

The solution detects underlying memory fault types with historical correctable errors (CE) information details. Instead of focusing on CE counting, the solution examines memory error data from multiple factors, such as spatial distribution (e.g., channel, rank, sub-channel, bank, row, column, etc.), temporal patterns (e.g., transient, intermittent, permanent), error bit location, memory-specific fault patterns, and failure risk according to host error correction code (ECC) design and configuration, etc.

▪ AI-assisted failure analysis

The solution leverages machine learning methods to train an AI model and predict memory failure by comparing thousands and thousands of memory logs on DDR5 from the field. The pre-trained memory failure prediction AI model is built in the baseboard management controller (BMC) and uses BMC to provide online prediction for memory failure, decreasing server downtime in large-scale datacenters.

▪ Integration with Alibaba Cloud Cruiser System

Real-time memory health assessment and predictive alerts are integrated with Alibaba Cloud Cruiser System, which provides fast and comprehensive hardware monitoring services for Alibaba Cloud's physical servers in datacenters.

Intel® Memory Resilience Technology (Intel® MRT)

Intel® Memory Resilience Technology enables data center operators to proactively predict potential memory failure risks, ensuring data center operation and workload continuity. This includes fine-grained OOB memory data collection, predictive failing DIMM alert, prediction-based memory page offline, prediction-based memory RAS actions, replacement of unreliable memory DIMMs, and critical workload migration.

By virtue of its multidimensional model and algorithms, Intel® Memory Resilience Technology examines DIMM errors at the micro-level to assign health scores and detect potential future failures in real-time. It utilizes AI to create a predictive by analyzing thousands of memory error logs from the field. By comparing this model to scans from the operator's data center, Intel® Memory Resilience Technology can pinpoint potential issues and provide an early detection to identify and prevent potential memory failures before they occur.

Diagnosing Hardware Issue with BDAT Data

Intel BIOS reference code implements system validation features that produce comprehensive system data, including memory margining data. The data is exposed from the standard BIOS Data ACPI table (BDAT), defined in the ACPI table. The BDAT data is fundamental support from system BIOS, which is produced throughout the BIOS boot flow and integrated into the ACPI RSDT

table. Analyzing the BDAT data is useful for diagnostics and issue debugging on production systems.

Results and Analysis

Alibaba Cloud has deployed Intel® Memory Resilience Technology on thousands of 4th Gen Intel® Xeon® Scalable processors in Alibaba Cloud datacenters across different workloads and is extending the deployment on 5th Gen Intel® Xeon® Scalable processors. The new generation processor has trusted performance and exceptional efficiency. It can achieve impressive performance per watt gains across all workloads, plus outsized performance and TCO in AI, database, networking, and HPC. 5th Gen Intel® Xeon® Scalable processors deliver more compute and faster memory at the same TDP as previous generation. It's software- and platform-compatible with previous generation, so the customers can minimize testing and validation when deploying new systems.

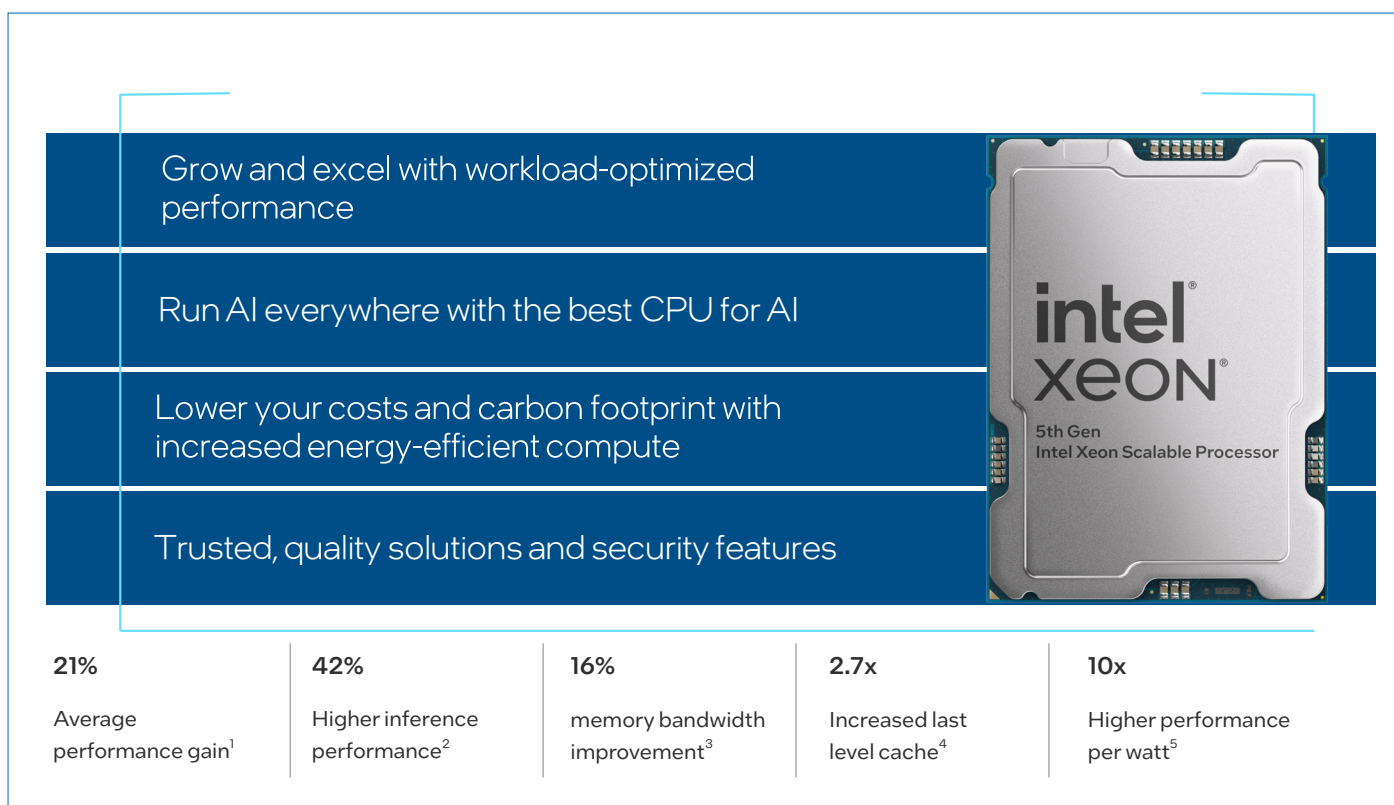


Figure 2. The 5th generation Intel® Xeon® Scalable processor has better performance.

¹ Average performance gain as measured by the geomean of SPEC CPU rate, STREAM Triad, and LINPACK compared to 4th Gen Intel® Xeon® processor. See G1 at intel.com/processorclaims: 5th Gen Intel Xeon Scalable processors. Results may vary.

² Based on performance gains of 1.1x to 1.29x for training (ResNet50v1.5, BERT-Large, SSD-ResNet34, RNN-T, MaskRCNN, and DLRM) and 1.19x to 1.42x for inference (ResNet50v1.5, BERT-Large, SSD-ResNet34, RNN-T (BF16 only), Resnext101 32x16d, MaskRCNN (BF16 only), DistilBERT) compared to 4th Gen Intel® Xeon® processor. See A15-A16 at intel.com/processorclaims: 5th Gen Intel Xeon Scalable processors. Results may vary.

³ See G12 at intel.com/processorclaims: 5th Gen Intel Xeon Scalable processors. Results may vary.

⁴ Based on Intel internal modelling as of December 2023.

⁵ Based on performance per watt gains of 1.46x to 10.6x with built-in accelerators on a range of AI, database, and networking workloads. See A19-A25, D1, D2, D5, N16 at intel.com/processorclaims: 5th Gen Intel Xeon Scalable processors. Results may vary.

The initial results indicate that the solution can effectively predict uncorrectable errors (UE) cases before they occur and alert correctable error (CE) storm cases before traditional CE counting based CE storm identification mechanism is triggered. The prediction leading time for UE and CE storm alert varies from minutes to hours and days depends underlying fault model. The

solution will undergo iterations and is expected to predict up to 57% UE and 74% CE storm with optimized DDR5 model⁶.

In addition to effective UE and CE storm prediction, the out-of-band (OOB) memory errors collected from BMC are critical for further diagnostic and troubleshooting of memory and system issues.

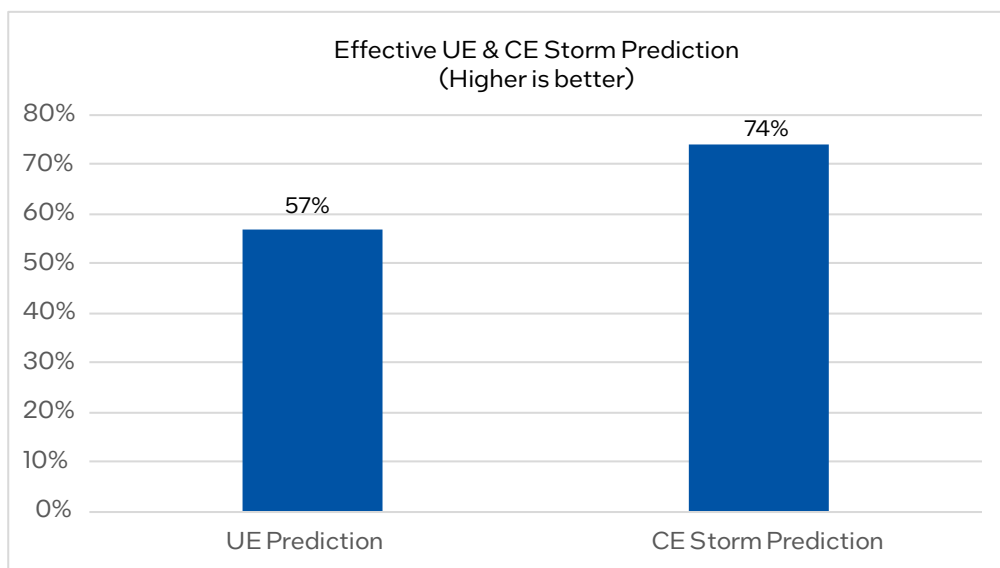


Figure 3. Effective UE & CE Storm Prediction

Conclusion

Alibaba Cloud has demonstrated the effectiveness of improving DDR5 memory reliability with AI-assisted prediction through BMC. It is crucial to improve overall data center total cost of ownership (TCO) for Alibaba Cloud. Intel® Memory Resilience Technology can detect the memory errors, alert imminent memory failure for uncorrectable errors (UE) and correctable errors (CE) storm, and improve memory reliability to keep the system running smoothly. Intel and Alibaba Cloud are collaborating on developing the methods to predict memory errors and retirement for DDR5 and new memory technologies.

⁶ Alibaba Cloud's internal test results as of November 2023. Test configuration: Intel® Xeon® 8475B processor and Intel® Xeon® 8575C processor, 32*64 GB, 16*64 GB, and 16*32 GB, 2*240 GB SSD. Intel does not control or audit third-party data. You should consult other sources to evaluate accuracy.

Author

Intel:

Shen Zhou (Data Center and AI Group/Intel China):

Principal Engineer in Intel Data Center and AI Group. Lead of Intel MRT technology. Seasoned expert in memory reliability, system RAS, and software-hardware co-design/optimization for memory architecture and cloud systems.

Zhijun Zheng (Data Center and AI Group/Intel China):

Cloud Software Engineer in Intel Data Center and AI. Technical support to Alibaba Cloud's server team firmware business. BIOS and BMC expert.

Alibaba Cloud:

Dahai Zhou - Houhai (Firmware Development/Alibaba Cloud Infrastructure Service Group):

BIOS technical experts. Responsible for the manageability, diagnosis, RAS, and other firmware feature design and development of cloud servers.

Zhibing Li – Bianhuang (Firmware Development/Alibaba Cloud Infrastructure Service Group):

BMC development manager. Lead of the BMC development of cloud servers.

Keke Xie – Xuanrui (Storage Component Development/Alibaba Cloud Infrastructure Service Group):

Memory technologist. Responsible for the development and management of memory components of cloud servers.



Performance varies by use, configuration and other factors. Learn more at [www.Intel.com/PerformanceIndex](https://www.intel.com/PerformanceIndex).

Performance results are based on testing as of dates shown in configurations and may not reflect all publicly available updates. See backup for configuration details. No product or component can be absolutely secure.

Your costs and results may vary.

Intel technologies may require enabled hardware, software or service activation.

Intel disclaims all express and implied warranties, including without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement, as well as any warranty arising from course of performance, course of dealing, or usage in trade.

Intel does not control or audit third-party data. You should consult other sources to evaluate accuracy.

© Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.